Final exam — Sample 2

DATA 412/612

Practice exam, questions only. Weeks 1–14. The newest material is lists, loops, and `purrr` (`pmap`, `keep`, `discard`). Earlier weeks stay in scope, including a report chunk.

Put your answers in an R Markdown document and keep the code visible except where a chunk option below says to hide it. You may use your notes and the course website. You may not use Google, any other website, or an LLM. Do not discuss the exam with anyone else. Load the tidyverse.

1. **A loop and a map** (8 pts). `bases` is `c(2, 3, 4)`. With a pre-allocated `for` loop, fill a numeric vector with the square of each entry. Then do the same job with `map_dbl()`. The two results should match.

2. `pmap` (6 pts). Using the same means and standard deviations as `mu = c(0, 10)` and `sigma = c(1, 2)`, draw 4 normal values from each pair with `pmap()` and a list of arguments. Set the seed to `612` first.

3. **`keep` and `discard`** (6 pts). From `mpg`, `keep` the numeric columns and report how many there are. Then `discard` every column that is not a character column, and print the names that remain.

4. **Tidy, then test** (12 pts). Reshape `wide` to a `year` column and a `sales` column. `year` is numeric. Drop the missing sales. Turn `store` into a factor ordered by total sales, largest first. Run a two-sample `t.test()` of `sales` for store `A` against store `B`. State the null hypothesis in one sentence.

```
wide <- tibble(
  store = c("A", "B", "A"),
  y2022 = c(5, 9, NA),
  y2023 = c(7, 4, 6)
)
```

5. **The report around that test** (8 pts). Put the test in a chunk labeled `store-test` with `echo = TRUE`. Put the reshape in a setup-style chunk with `include = FALSE`. Send the store totals to `knitr::kable()` with a caption. In YAML, set a bibliography file `refs.bib`, and in the text cite `wickham2017r4ds` in parentheses.